

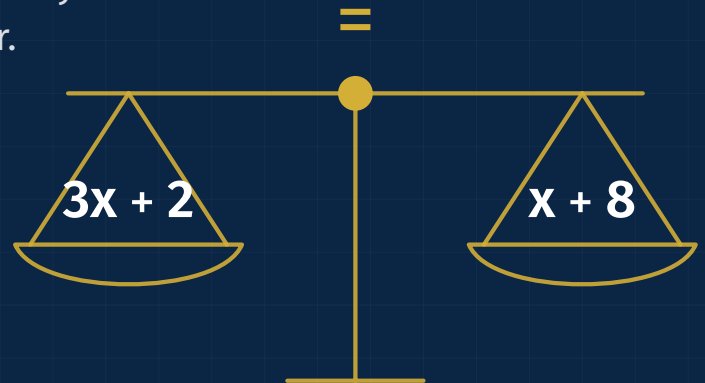


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CHAPTER 2 • ALGEBRA • TOPIC 3

Expanding Brackets

Multiply every term by every term — one bracket, two, three. The grid keeps every product, and CALC checks the answer.



7

VARIANTS

19

PAST-PAPER PARTS

51

PRACTICE QUESTIONS

YOU WILL MASTER

V1 One bracket

V2 Expand & simplify

V3 Two brackets

V4 Squares & DOTS

V5 Three brackets

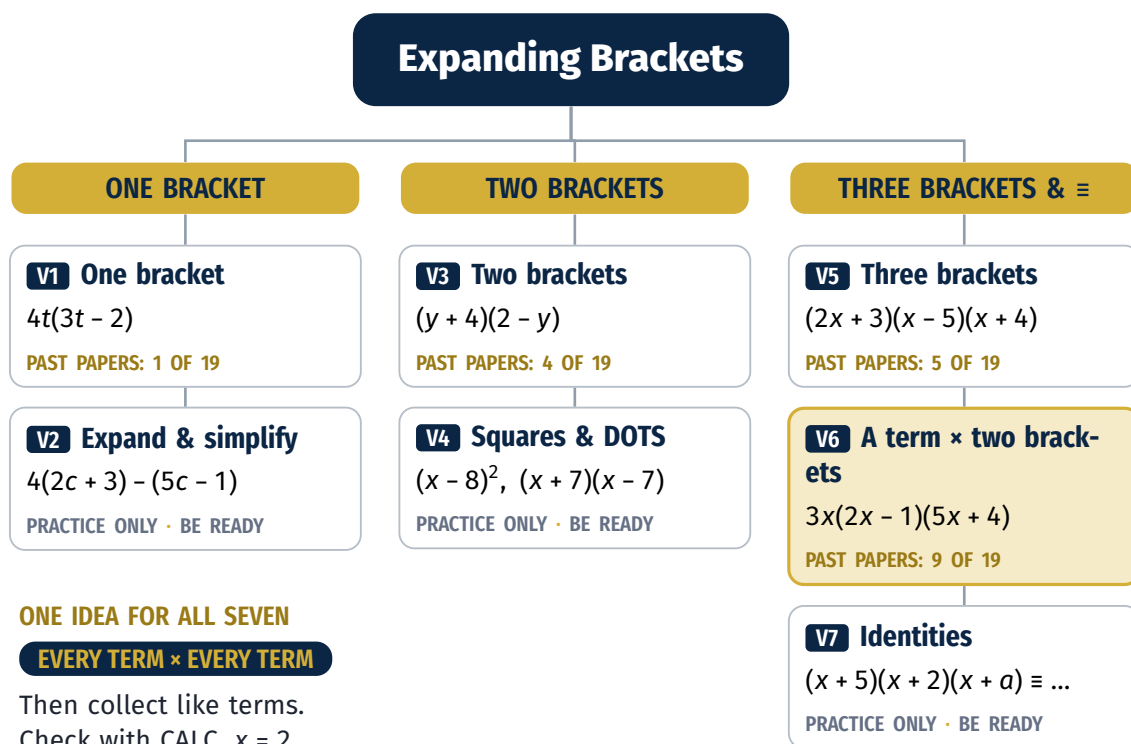
V6 A term $\times$ two brackets

V7 Identities

TOPIC MAP · EVERY VARIANT ON ONE PAGE

Expanding Brackets at a glance



Your notes

 How often does each variant come up?

18 past-paper questions (Oct/Nov 2020 → Nov 2025) hold 19 expanding parts; 51 practice parts — grouped by type.

	PAST PAPERS (OUT OF 19)	PRACTICE PARTS
V1 One bracket	1	12
V2 Expand & simplify	0	9
V3 Two brackets	4	14
V4 Squares & DOTS	0	2
V5 Three brackets	5	9
V6 A term × two brackets	9	2
V7 Identities	0	3

WHAT THIS TELLS YOU 14 of the 19 past-paper parts are **three** factors (V5 + V6), 3 marks each. Two brackets (V3) is the warm-up at the start of the paper.

BEFORE YOU START

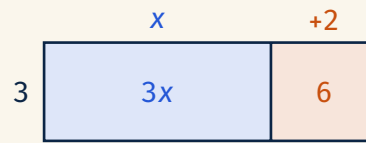
The big idea



Your notes
Expanding = multiplying out.

A bracket is a rectangle: $3(x + 2)$ is a rectangle 3 by $(x + 2)$. Its area is $3x + 6$ — every part of the length times the width.

Expand and **factorise** are opposites (Topic 4 goes backwards).



$$3(x + 2) \xrightarrow{\text{EXPAND}} 3x + 6$$

$$3x + 6 \xleftarrow{\text{FACTORISE}} 3(x + 2)$$

Sign rules — every product needs a sign

$$+ \times + = +$$

$$3 \times 2 = 6$$

same signs →
plus

$$- \times - = +$$

$$-2 \times (-5) = +10$$

same signs →
plus

$$+ \times - = -$$

$$4 \times (-x) = -4x$$

different → minus

$$- \times + = -$$

$$-3 \times 2x = -6x$$

different → minus

Letters: $x \times x = x^2$, $x \times x^2 = x^3$, $3x \times 4x = 12x^2$ (numbers × numbers, letters × letters).

How many products?
ONE BRACKET

$a(b + c)$: 2 products

TWO BRACKETS

2×2 grid: 4 products

THREE BRACKETS

two first, then the third

**CALCULATOR FIRST, THEN CHECK**

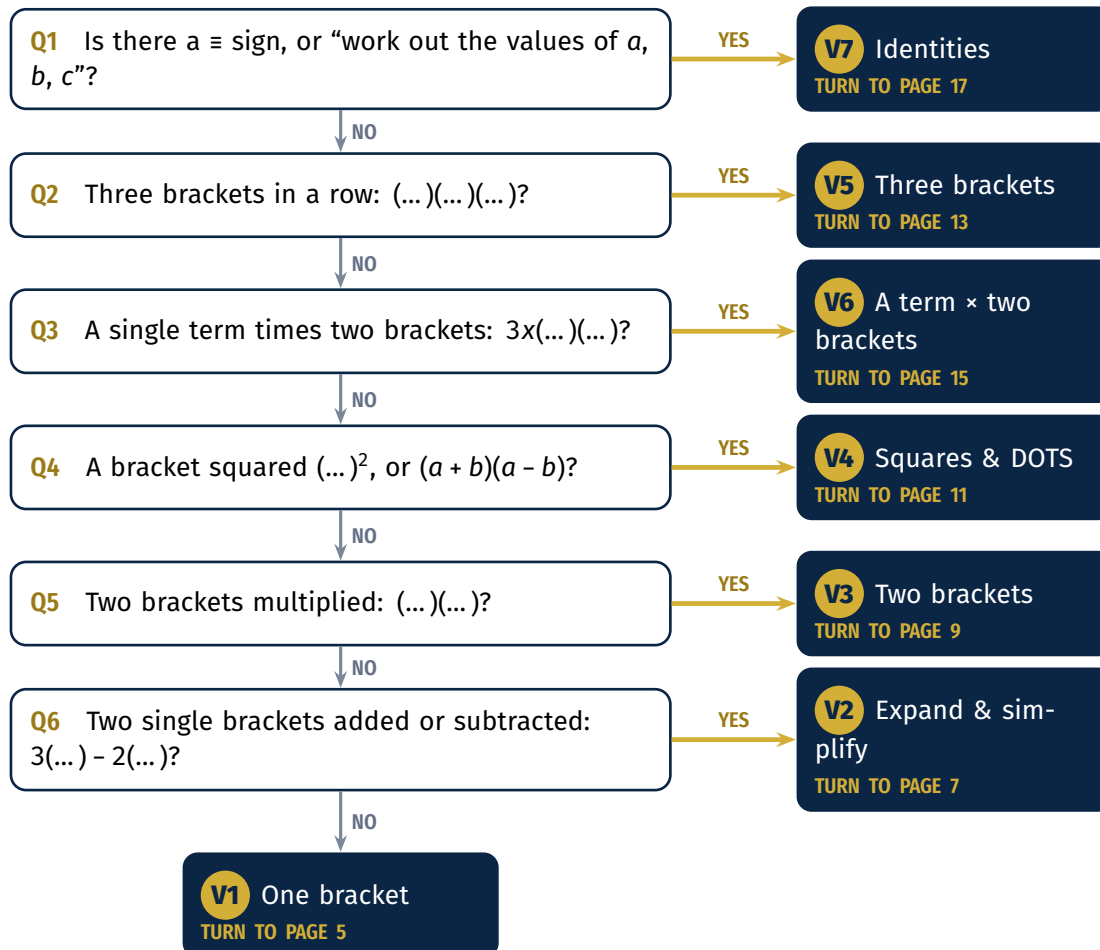
Check every expansion with **CALC** (Dr Eslam's habit). Type the question, press **CALC**, $x = 2$, **=**. Then type your answer, **CALC**, $x = 2$. The same number? Your expansion is right. Different? Look for a lost sign or a lost product.

COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?

 Your notes

Count the brackets, then walk down the ladder. The first **YES** tells you the variant – and the page to turn to.



COLOUR CODE IN THE WORKING

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD

- | | |
|---|---|
|  What is it? |  When you see → you do |
|  How do I do it? |  Watch out |
|  Worked example |  Exam tip |
|  Check yourself |  Calculator check |

V1

VARIANT 1

One Bracket

A term outside multiplies every term inside.

SEEN IN PAST PAPERS

1 / 19

PRACTICE QUESTIONS: 12



Your notes

Came up in: Jan 2021 P2HR Q1(c)

? What is it?**In plain words**

The term outside the bracket multiplies **every term** inside. Two terms inside give two answers.

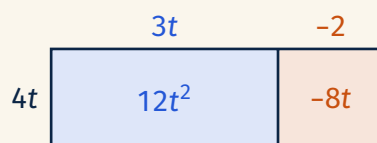
If the outside term has a letter, the powers add: $4t \times 3t = 12t^2$.

IT LOOKS LIKE

$3(2y - 5)$

$2m(m + 3)$

$4x^2(3x + 5)$



$$4t(3t - 2)$$

Area model: $4t$ by $(3t - 2)$ — two pieces.**How do I do it?****1 Arrows**

Draw an arrow from the outside term to **each** term inside.

2 Multiply

Numbers $\times$ numbers, letters $\times$ letters: $4t \times 3t = 12t^2$.

3 Signs

Keep the sign with each term: $4t \times (-2) = -8t$.

OUTSIDE $\times$ EVERY TERM INSIDE

Two terms inside $\rightarrow$ two terms out.

V1 One Bracket

VARIANT 1 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Expand” or “multiply out” → **Outside term × every inside term**

A letter outside: $x(x - 3)$, $3y(4y - 3)$ → $x \times x = x^2$: **powers add**

“Circle your answer” (1 mark) → **Expand on the side first, then circle**

**WORKED EXAMPLE****PAST PAPER**

January 2021 · Paper 2HR · Q1(c) · 1 mark

Expand $4t(3t - 2)$

STEP 1 $4t \times 3t$ and $4t \times (-2)$ ◀ arrows to both terms

STEP 2 $12t^2$ ◀ $4 \times 3 = 12$, $t \times t = t^2$

STEP 3 $-8t$ ◀ plus times minus is minus

ANSWER $12t^2 - 8t$ ✓ CALC $t = 2$: both sides 32**WATCH OUT**

$$\times 4t(3t - 2) = 12t^2 - 2$$

$$\checkmark 4t(3t - 2) = 12t^2 - 8t$$

The $4t$ multiplies the -2 as well — every term inside.

**EXAM TIP**

A 1-mark expand is quick — but still check: put $t = 2$ into both forms.

**Check yourself**

calculator first · answers on page 19

1.1

Practice · Easy

Expand $3y(4y - 3)$

Answer:

1.2

Practice · Easy

Expand $4x^2(3x + 5)$

Answer:

1.3

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Expand $-3(2x - 5)$

Answer:

V2

VARIANT 2

Expand & Simplify

Two single brackets added or taken away —
the minus is the trap.

SEEN IN PAST PAPERS

0 / 19

NOT ASKED YET · BE READY
PRACTICE QUESTIONS: 9


Your notes
? What is it?**In plain words**

Expand each bracket on its own, then
collect like terms.

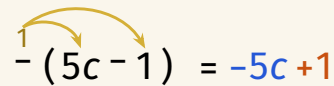
A **minus in front** of a bracket is a hidden
-1: it changes **every** sign inside:
 $-(5c - 1) = -5c + 1$.

IT LOOKS LIKE

$$2(x - 4) + 3(x + 5)$$

$$3(m + 4) - 2(4m + 1)$$

$$4(2c + 3) - (5c - 1)$$



$$-(5c - 1) = -5c + 1$$

THE HIDDEN -1 HITS BOTH TERMS**How do I do it?****1 Expand each bracket**

One bracket at a time. A minus in front: multiply by -1 (or -2, ...).

2 Collect

Letter terms together, numbers together.

3 Tidy

Letter term first, number last: $3c + 13$.

MINUS IN FRONT CHANGES EVERY SIGN $-(5c - 1) = -5c + 1$.

V2 Expand & Simplify

VARIANT 2 · SEE IT DONE, THEN TRY


 Your notes
WHEN YOU SEE → YOU DO*"Expand and simplify"*→ **Expand every bracket, then collect like terms** $-(\dots)$ or $-2(\dots)$ → **The minus changes every sign inside** $3x(2x + 3) - x(3x + 5)$ → **Same move — the x^2 terms collect too****WORKED EXAMPLE****PRACTICE**

Expanding Brackets · Medium Q16 · 2 marks

Expand and simplify fully $4(2c + 3) - (5c - 1)$ **STEP 1**

$$8c + 12 - 5c + 1$$

◀ expand both; the minus flips -1 to $+1$ **STEP 2**

$$8c - 5c + 12 + 1$$

◀ collect like terms

STEP 3

$$3c + 13$$

◀ tidy

ANSWER $3c + 13$ ✓ CALC $c = 2$: both sides 19**WATCH OUT**

$$\times -(5c - 1) = -5c - 1$$

$$\checkmark -(5c - 1) = -5c + 1$$

 $-1 \times -1 = +1$. The minus in front hits the second term too.**EXAM TIP**Put a bracket around the hidden number: $-1(5c - 1)$. Now it looks like V1.**Check yourself**

calculator first · answers on page 19

2.1

Practice · Easy

Expand and simplify
 $3(y - 2) + 5(2y + 1)$

Answer:

2.2

Practice · Medium

Expand and simplify
 $3(m + 4) - 2(4m + 1)$

Answer:

2.3

Practice · Medium

Expand and simplify
 $3x(2x + 3) - x(3x + 5)$

Answer:

V3

VARIANT 3

Two Brackets

Four products — the grid makes sure you get them all.

SEEN IN PAST PAPERS

4 / 19

PRACTICE QUESTIONS: 14



Your notes

Came up in: Nov 2021 P1H Q1(b) • Jan 2021 P2HR Q1(d) • Jan 2022 P2H Q1(a) • Jun 2023 P2HR Q1(b)

? What is it?**In plain words**

Every term of the first bracket multiplies every term of the second: $2 \times 2 = 4$ **products**.

Put them in a **grid**. The two middle ones are usually like terms — collect them.

IT LOOKS LIKE

$$(x + 5)(x - 7)$$

$$(5x - 2)(x + 4)$$

$$(y + 4)(2 - y)$$

×	2	-y
y	2y	-y ²
+4	8	-4y

$$-y^2 + \underbrace{2y - 4y}_{\text{collect}} + 8 = 8 - 2y - y^2$$

How do I do it?**1 Draw the grid**

First bracket down the side, second along the top — **signs stay with their terms**.

2 Fill 4 cells

Row × column in every cell.

3 Collect

The like terms (usually the diagonal) add up.

4 Write it

In order: x^2 term, x term, number.

THE GRID Four cells, four products, nothing lost.

V3 Two Brackets

VARIANT 3 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Expand and simplify $(x + 5)(x - 7)$ ” → **Grid: 4 products, collect the middle**

A bracket written backwards: $(2 - y)$ → **Keep its order and signs: 2 and $-y$**

“Circle the term which is part of the answer” → **Only work out the middle terms**

WORKED EXAMPLE **PAST PAPER**

January 2022 · Paper 2H · Q1(a) · 2 marks

Expand and simplify $(y + 4)(2 - y)$

STEP 1 grid: y , $+4$ down the side; 2 , $-y$ along the top ◀ signs stay with their terms

STEP 2 $2y - y^2 + 8 - 4y$ ◀ four products

STEP 3 $-y^2 + (2y - 4y) + 8$ ◀ collect the like terms

STEP 4 $8 - 2y - y^2$ ◀ tidy

ANSWER $8 - 2y - y^2$ ✓ CALC $y = 2$: both sides 0**WATCH OUT**

$$\times (y + 4)(2 - y) = y^2 - 2y + 8$$

$$\checkmark (y + 4)(2 - y) = 8 - 2y - y^2$$

$y \times (-y) = -y^2$: the minus belongs to the y in the second bracket.

EXAM TIP

At the start of Paper 1 and 2 — 2 quick marks. Write the 4 products before collecting: if a sign slips you still get M1.

Check yourself

calculator first · answers on page 19

3.1 Jun 2023 P2HRExpand and simplify
 $(x + 5)(x - 7)$

Answer:

3.2 Jan 2021 P2HRExpand and simplify
 $(5x - 2)(x + 4)$

Answer:

3.3 Practice · Easy

$(2x - 4)(3x + 5)$ is expanded. Which is in the answer: $2x$, $-2x$, $22x$ or $-22x$?

Answer:

V4

VARIANT 4

Squares & DOTS

A squared bracket has a middle term;
 $(a + b)(a - b)$ has none.

SEEN IN PAST PAPERS

0 / 19

NOT ASKED YET • BE READY
 PRACTICE QUESTIONS: 2


 Your notes
? What is it?**In plain words**

$(x + 4)^2$ means $(x + 4)(x + 4)$ —

four pieces, not two. Quick rule:
square the first, double the product,
square the last.

$(x + 7)(x - 7)$: same numbers, opposite
 signs — the middle terms **cancel**:
 $x^2 - 49$ (Difference Of Two Squares).

IT LOOKS LIKE

$$(x - 8)^2$$

$$(w - 5)^2$$

$$(2x - 5)(2x + 5)$$

	x	4
4	$4x$	16
x	x^2	$4x$

$$(x + 4)^2 = x^2 + 8x + 16$$

SQUARE • DOUBLE • SQUARE

How do I do it?**1 Square → two brackets**

Write $(x - 8)^2$ as $(x - 8)(x - 8)$ — or use the rule.

2 Square, double, square

$$x^2, 2 \times x \times (-8) = -16x, (-8)^2 = +64.$$

3 DOTS

$(a + b)(a - b)$: the middle terms cancel, answer $a^2 - b^2$.

SQUARE • DOUBLE • SQUARE $(a + b)^2 = a^2 + 2ab + b^2$.

V4 Squares & DOTS

VARIANT 4 · SEE IT DONE, THEN TRY


 Your notes

WHEN YOU SEE → YOU DO

"Multiply out $(x - 8)^2$ " → **Square · double · square:** $x^2 - 16x + 64$

$(2x + 5)(2x - 5)$: same numbers, opposite signs → **Middle cancels:** $4x^2 - 25$

Completing the square (Topic 5) → **You will need this rule backwards**



WORKED EXAMPLE

PRACTICE

Expanding Brackets · Easy Q16 · 2 marks

Multiply out and simplify $(x - 8)^2$

STEP 1 $(x - 8)(x - 8)$ ◀ a square is the bracket twice

STEP 2 $x^2 - 8x - 8x + 64$ ◀ four products

STEP 3 $x^2 - 16x + 64$ ◀ collect: double the product

ANSWER $x^2 - 16x + 64$

✓ CALC $x = 2$: both sides 36



WATCH OUT

$$\times (x - 8)^2 = x^2 - 64$$

$$\checkmark (x - 8)^2 = x^2 - 16x + 64$$

A squared bracket keeps its middle term.
And $(-8)^2 = +64$.



EXAM TIP

Only $(a + b)(a - b)$ loses its middle —
check the signs before you use DOTS.



Check yourself

calculator first · answers on page 19

4.1

Practice · Medium

Expand and simplify
 $(w - 5)^2$

Answer:

4.2

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Expand $(2x + 3)^2$

Answer:

4.3

Dr Eslam

Expand $(2x - 5)(2x + 5)$

Answer:

V5

VARIANT 5

Three Brackets

Two brackets first, then multiply by the third.

SEEN IN PAST PAPERS

5 / 19

PRACTICE QUESTIONS: 9



Your notes

Came up in: Oct/Nov 2020 P1HR Q14 · Jan 2021 P1HR Q16(a) · Jun 2022 P1HR Q16(a) · Nov 2023 P2H Q16 · May 2024 P1HR Q14(a)

? What is it?**IN PLAIN WORDS**

Three brackets is just two jobs: multiply **two brackets first** (a 2×2 grid, 3 terms out), then multiply that answer by the **third bracket** (a 3×2 grid). The answer has an x^3 term: $ax^3 + bx^2 + cx + d$.

IT LOOKS LIKE

$(2x + 3)(x - 5)(x + 4)$

$(x + 4)(x - 2)(x + 1)$

$(3x + 1)(2 - x)(4 + x)$

$(2x + 3)(x - 5) \xrightarrow{\text{JOB 1}} 2x^2 - 7x - 15$

JOB 2: $\times (x + 4)$

	$2x^2$	$-7x$	-15
$\times x$	$2x^3$	$-7x^2$	$-15x$
$+4$	$8x^2$	$-28x$	-60

$$2x^3 + (8 - 7)x^2 + (-28 - 15)x - 60 = 2x^3 + x^2 - 43x - 60$$

Job 1: two brackets $\rightarrow$ 3 terms. Job 2: a 3×2 grid with the third bracket.**How do I do it?****1 Two brackets**

Grid the first two brackets: 4 products.

2 SimplifyCollect to 3 terms: $2x^2 - 7x - 15$.**3 $\times$ the third**A 3×2 grid with the third bracket: 6 products.**4 Collect** x^3 , x^2 , x , number — in that order.**TWO FIRST, THEN THE THIRD**

Write the 3-term line — it is a method mark.

V5 Three Brackets

VARIANT 5 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO“Expand and simplify $(...)(...)(...)$ ”→ **Two brackets first, then the third**“Show that ... can be written as $ax^3 + bx^2 + cx + d$ ”→ **Expand fully; say what a, b, c, d are**

(3 marks)

→ **M1 two brackets, M1 the third, A1 answer****WORKED EXAMPLE****PAST PAPER**

November 2023 · Paper 2H · Q16 · 3 marks

Expand and simplify $(2x + 3)(x - 5)(x + 4)$ **STEP 1** $(2x + 3)(x - 5) = 2x^2 - 10x + 3x - 15$ ◀ two brackets first**STEP 2** $= 2x^2 - 7x - 15$ ◀ simplify to 3 terms**STEP 3** $2x^3 + 8x^2 - 7x^2 - 28x - 15x - 60$ ◀ $\times(x + 4)$: six products**STEP 4** $2x^3 + x^2 - 43x - 60$ ◀ collect**ANSWER** $2x^3 + x^2 - 43x - 60$ ✓ CALC $x = 2$: both sides -126**WATCH OUT**

× forgetting a product:

$$2x^3 + 8x^2 - 7x^2 - 15x - 60$$

✓ 3×2 grid: six products, $-28x$ includedA 3×2 grid has 6 cells — count your products before you collect.**EXAM TIP**CALC with $x = 2$ catches almost every slip in a three-bracket expansion — 30 seconds well spent.**Check yourself**

calculator first · answers on page 19

5.1

Jan 2021 P1HR

Expand and simplify
 $(x + 4)(x - 2)(x + 1)$

Answer:

5.2

Oct/Nov 2020 P1HR

Expand and simplify
 $(4x + 1)(x - 3)(5x + 6)$

Answer:

5.3

May 2024 P1HR

Expand and simplify
 $(3x + 1)(2 - x)(4 + x)$

Answer:

V6**VARIANT 6** MOST COMMON IN THE EXAM

SEEN IN PAST PAPERS

9 / 19

PRACTICE QUESTIONS: 2

A Term × Two Brackets

A single term in front of two brackets:
brackets first, the term last.

Came up in: Jan 2021 P1H Q12(b) • May 2021 P1H Q13 • Nov 2021 P2H Q12(c) • Jan 2022 P1H Q13(a) • Jun 2024 P2H Q14(a) • Nov 2024 P1H Q15 • Jun 2025 P2H Q12 • Jun 2025 P2HR Q13(b) • Nov 2025 P1H Q14(a)


Your notes

? What is it?

IN PLAIN WORDS

A term like $3x$ sits in front of two brackets. Multiply the **two brackets first**, simplify, then multiply **every term** by $3x$. There is no number term at the end — the answer is $ax^3 + bx^2 + cx$.

IT LOOKS LIKE

$3x(2x - 1)(5x + 4)$

$4x(3x + 1)(2x - 3)$

$2x(x - 5)(x - 3)$

1 TWO BRACKETS

$(2x - 1)(5x + 4)$

2 SIMPLIFY

$10x^2 + 3x - 4$

3 × THE TERM

$3x \times (10x^2 + 3x - 4)$

$30x^3 + 9x^2 - 12x$

Brackets first — the single term goes last, so its letter only touches 3 terms.

How do I do it?

1 Two brackets

Grid the two brackets: 4 products.

2 Simplify

Collect to 3 terms.

3 × the term

Multiply each of the 3 terms by the single term (powers add).

4 Form

Write it as $ax^3 + bx^2 + cx$ and state a , b , c if asked.

BRACKETS FIRST, TERM LAST

The most common expanding question.

V6 A Term × Two Brackets

VARIANT 6 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Expand and simplify $3x(2x + 5)(7x - 4)$ ” → **Brackets first, then $\times 3x$**

“Write ... in the form $ax^3 + bx^2 + cx$ ” → **Expand fully; read off a, b, c**

“Show your working clearly” → **Write the 3-term line before multiplying by $3x$**

WORKED EXAMPLE **PAST PAPER**

November 2024 · Paper 1H · Q15 · 3 marks

Write $3x(2x - 1)(5x + 4)$ in the form $ax^3 + bx^2 + cx$ where a, b and c are integers.

STEP 1 $(2x - 1)(5x + 4) = 10x^2 + 8x - 5x - 4$ ◀ two brackets first

STEP 2 $= 10x^2 + 3x - 4$ ◀ simplify

STEP 3 $3x \times 10x^2 + 3x \times 3x - 3x \times 4$ ◀ every term $\times 3x$

STEP 4 $30x^3 + 9x^2 - 12x$ ◀ $a = 30, b = 9, c = -12$

ANSWER $30x^3 + 9x^2 - 12x$ ✓ CALC $x = 2$: both sides 252**WATCH OUT**

$$\times 3x(2x - 1) = 6x - 1$$

$$\checkmark 3x(2x - 1) = 6x^2 - 3x$$

If you start with the term, it must multiply **both** terms — another reason to do the brackets first.

EXAM TIP

9 of the 19 past-paper parts are this variant (3 marks). Three clear lines: grid, simplify, $\times$ the term.

Check yourself

calculator first · answers on page 19

6.1 Jan 2021 P1H

Expand and simplify
 $2x(x - 5)(x - 3)$

Answer:

6.2 May 2021 P1H

Expand and simplify
 $4x(3x + 1)(2x - 3)$

Answer:

6.3 Jun 2024 P2H

Write $3y(2y + 5)(y + 7)$
as $ay^3 + by^2 + cy$

Answer:

V7

VARIANT 7

Identities

$\equiv$: expand, then make the coefficients match.

SEEN IN PAST PAPERS

0 / 19

NOT ASKED YET • BE READY
PRACTICE QUESTIONS: 3



Your notes

? What is it?

In plain words

An **identity** ($\equiv$) is true for **every** x .
So after expanding, the x^3 parts match, the x^2 parts match, the x parts match and the numbers match.

“Show that” questions are the same: expand, then rearrange to what they want.

IT LOOKS LIKE

$$a(3x - 10) \equiv 21x + 2b$$

$$(x + 5)(x + 2)(x + a) \equiv \dots$$

$$\begin{array}{ccc} x^3 & +4x^2 & -11x & -30 \\ \updownarrow & & & \\ x^3 & +bx^2 & +cx & -30 \end{array} \quad \equiv$$

$b = 4, c = -11$

How do I do it?

1 Expand fully

Expand the side with brackets (letters like a stay as letters).

2 Line up

Write both sides in the same order: x^3 , x^2 , x , number.

3 Match

Same power on both sides: their coefficients are equal.

4 Solve

Solve each small equation. Use the easy one (the numbers) first.

MATCH THE COEFFICIENTS Start with the numbers: $5 \times 2 \times a = -30$.

V7 Identities

VARIANT 7 · SEE IT DONE, THEN TRY

 Your notes

WHEN YOU SEE → YOU DO

 $\equiv$, "work out the values of a , b and c " → Expand, then match coefficients

 "Show that ... for all values of x " → Expand the left side until it equals the right

 "Show that $x^2 + 6x = 1$ " (area) → Form the expression, expand, rearrange

WORKED EXAMPLE PRACTICE

Expanding Brackets · Hard Q14 · 3 marks

 $(x + 5)(x + 2)(x + a) \equiv x^3 + bx^2 + cx - 30$. Work out the values of the integers a , b and c .

STEP 1 $5 \times 2 \times a = -30 \Rightarrow a = -3$ ◀ numbers match first

STEP 2 $(x + 5)(x + 2) = x^2 + 7x + 10$ ◀ two brackets

STEP 3 $(x^2 + 7x + 10)(x - 3) = x^3 + 4x^2 - 11x - 30$ ◀ $\times$ the third

STEP 4 $b = 4, c = -11$ ◀ match x^2 and x
ANSWER $a = -3, b = 4, c = -11$
✓ CALC $x = 2$: both sides -28

WATCH OUT

 $\times a(3x - 10) \equiv 21x + 2b \Rightarrow 3a = 21, -10 = 2b$
 $\checkmark 3a = 21$ and $-10a = 2b$

 The a multiplies the -10 too. Then $a = 7$, $b = -35$.

EXAM TIP

 An identity works for every x — test your a , b , c with $x = 1$ or $x = 2$ on both sides.

Check yourself

calculator first · answers on page 19

7.1 Practice · Hard

 $a(3x - 10) \equiv 21x + 2b$.
Work out a and b .

Answer:

7.2 Dr Eslam

 Show that
 $(n + 3)^2 - (n - 3)^2 = 12n$
for all n .

Answer:

7.3 Practice · Hard

 A square has side $(3 + x)$
cm and area 10 cm^2 .
Show that $x^2 + 6x = 1$.

Answer:

CHECK YOURSELF

Answers

Every answer below was checked with sympy (check_answers.py). Got one wrong? Go back to that variant's Watch-out box first.

1.1 $12y^2 - 9y$

1.2 $12x^3 + 20x^2$

1.3 $-6x + 15$

2.1 $13y - 1$

2.2 $10 - 5m$

2.3 $3x^2 + 4x$

3.1 $x^2 - 2x - 35$

3.2 $5x^2 + 18x - 8$

3.3 $-2x$

4.1 $w^2 - 10w + 25$

4.2 $4x^2 + 12x + 9$

4.3 $4x^2 - 25$

5.1 $x^3 + 3x^2 - 6x - 8$

5.2
 $20x^3 - 31x^2 - 81x - 18$

5.3 $-3x^3 - 7x^2 + 22x + 8$

6.1 $2x^3 - 16x^2 + 30x$

6.2 $24x^3 - 28x^2 - 12x$

6.3 $6y^3 + 57y^2 + 105y$

7.1 $a = 7, b = -35$

7.2
 $n^2 + 6n + 9 - (n^2 - 6n + 9) = 12n$

7.3 $(3 + x)^2 = 10 \Rightarrow x^2 + 6x + 9 = 10 \Rightarrow x^2 + 6x = 1$

THE WHOLE TOPIC ON ONE PAGE

Expanding Brackets cheat sheet



Your notes

V1 One bracket 1/19

$$4t(3t - 2) = 12t^2 - 8t$$

Outside $\times$ every term inside.**V2 Expand & simplify**

$$4(2c + 3) - (5c - 1) = 3c + 13$$

Minus in front changes every sign.

V3 Two brackets 4/19

$$(y + 4)(2 - y) = 8 - 2y - y^2$$

The grid: 4 products, collect the middle.

V4 Squares & DOTS

$$(x - 8)^2 = x^2 - 16x + 64$$

Square $\cdot$ double $\cdot$ square.

$$(a + b)(a - b) = a^2 - b^2.$$

V5 Three brackets 5/19

$$(2x + 3)(x - 5)(x + 4)$$

Two first (3 terms), then a 3×2 grid.**V6 Term $\times$ two brackets 9/19**

$$3x(2x - 1)(5x + 4) = 30x^3 + 9x^2 - 12x$$

Brackets first, the term last.

V7 Identities

$$x^3 + bx^2 + cx - 30$$

Expand, line up, match the coefficients.

**TOP 3 LOST MARKS**

1. $-(5c - 1) = -5c - 1 \times$

2. $(x - 8)^2 = x^2 - 64 \times$

3. a product missing from the grid $\times$ **BOTTOM LINE** Every term $\times$ every term $\cdot$ the grid $\cdot$ two brackets first $\cdot$ CALC $x = 2$ to check.